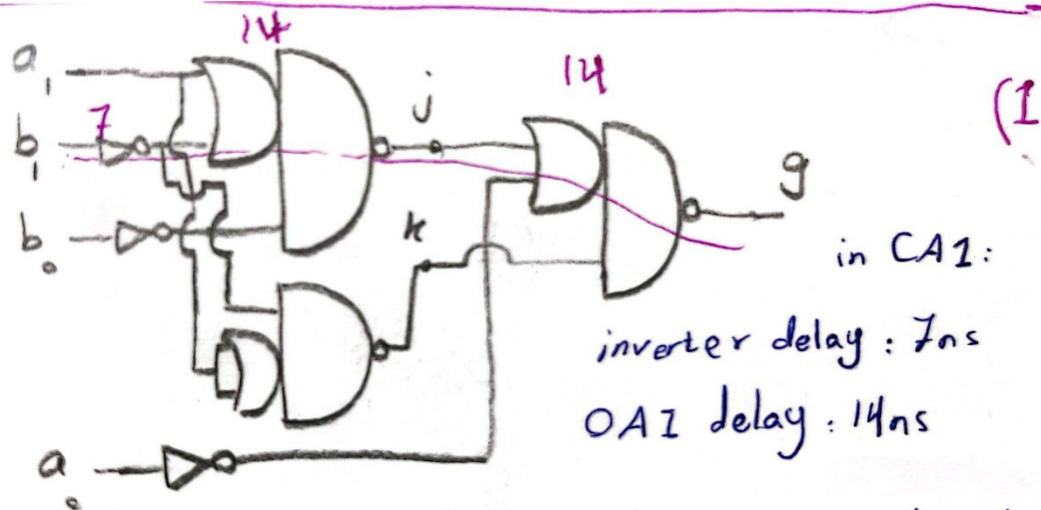
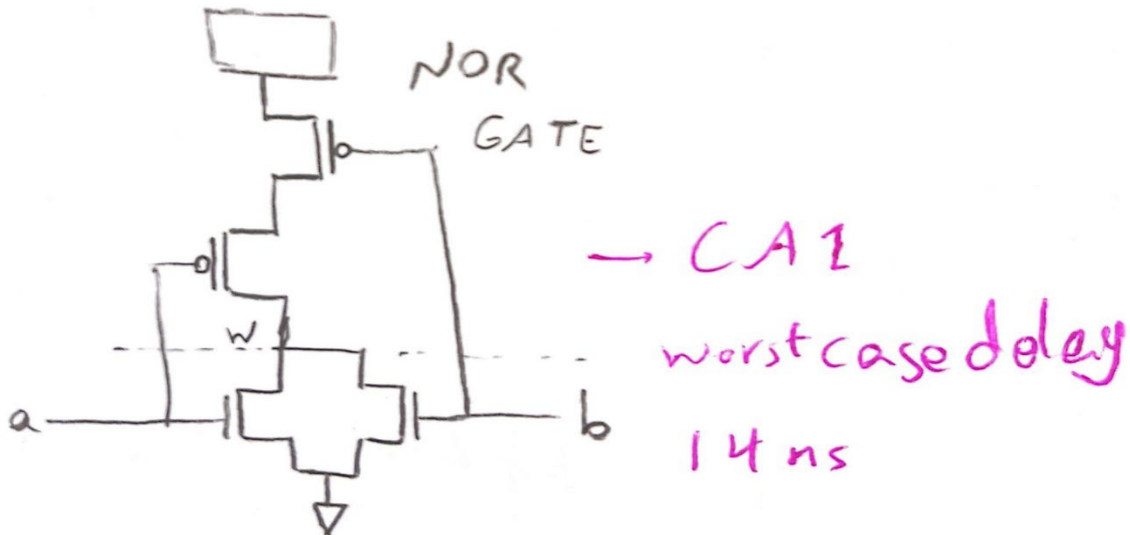


IN THE NAME OF GOD

## CA2 REPORT

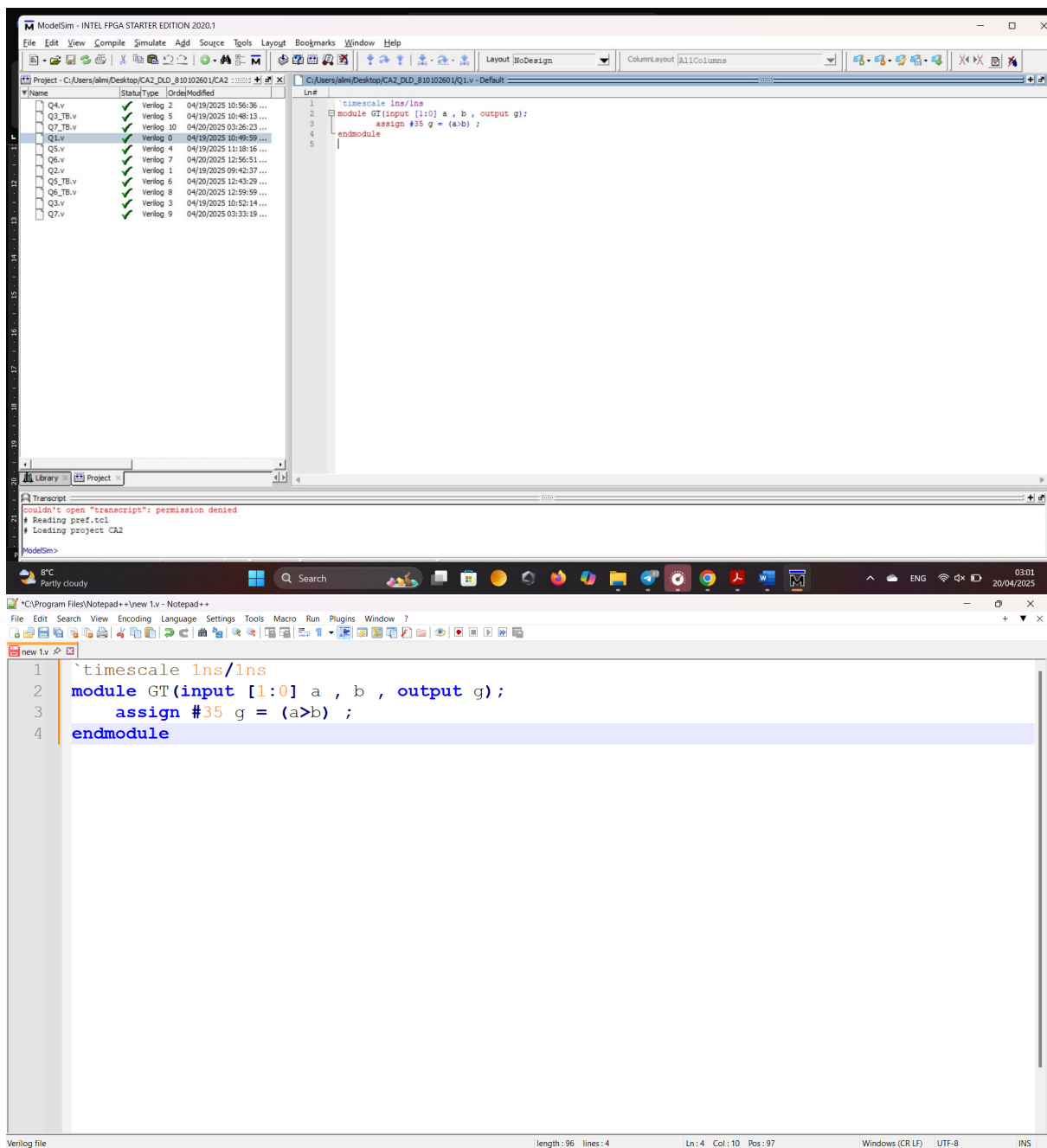
Fullname: Amirmohammad Mirhosseini , SID:810102601 , Logic No: 92

Q1)

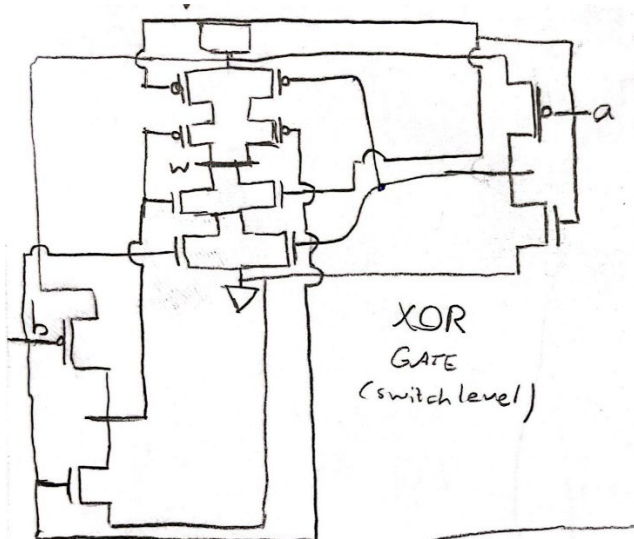


Worst case delay happens in selected path:

$$7 + 14 + 14 = 35 \text{ ns}$$



Q2)



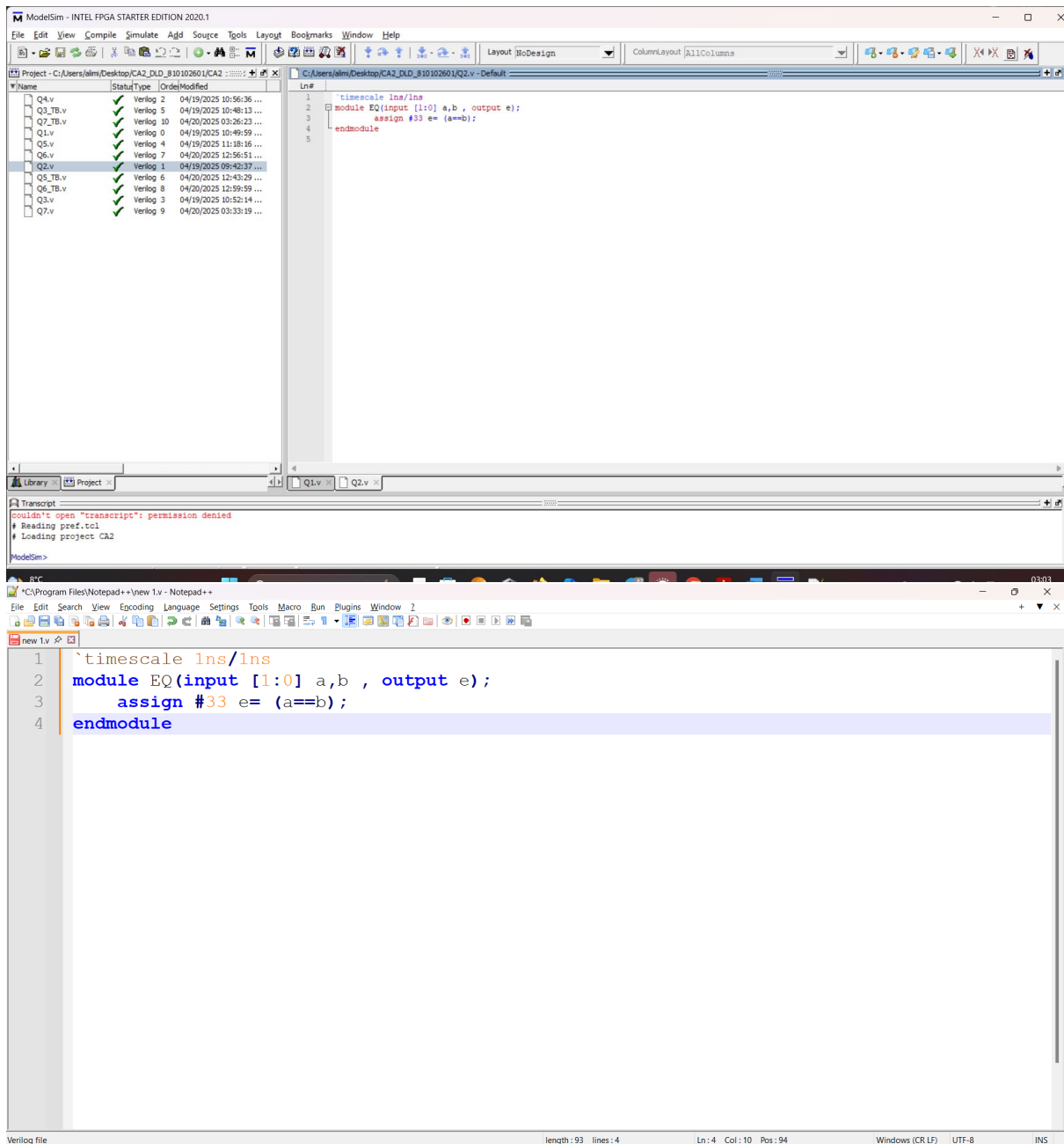
finding XOR gate worst case delay

(2)

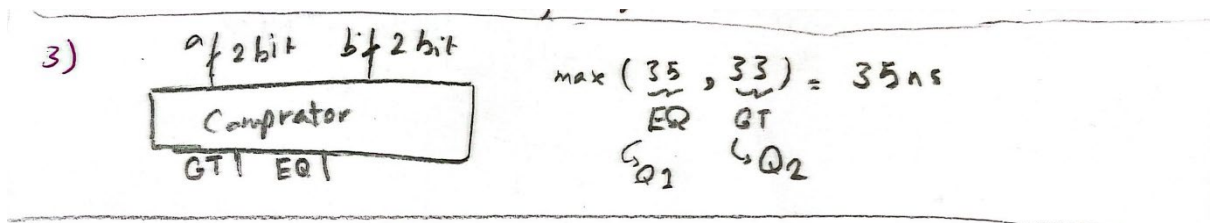
$$t_{00} : \max(5+7+7, 4+4+5) \rightarrow$$

$$t_{01} : \max(5+5+5, 5+5+5) \rightarrow \text{max} \rightarrow 19\text{ns}$$

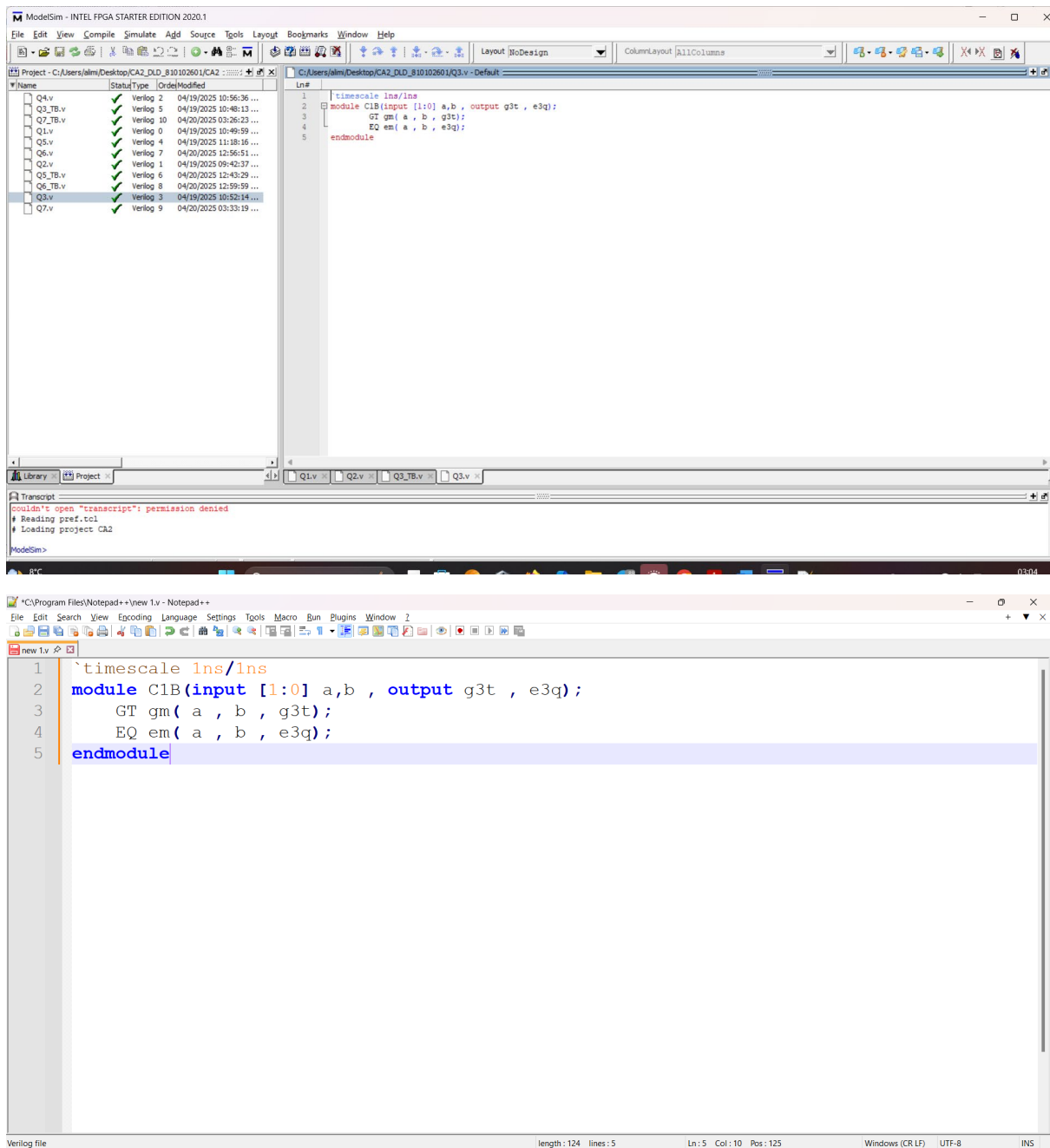
$$\text{worst case delay} : 19 + 14 = \boxed{33\text{ns}}$$



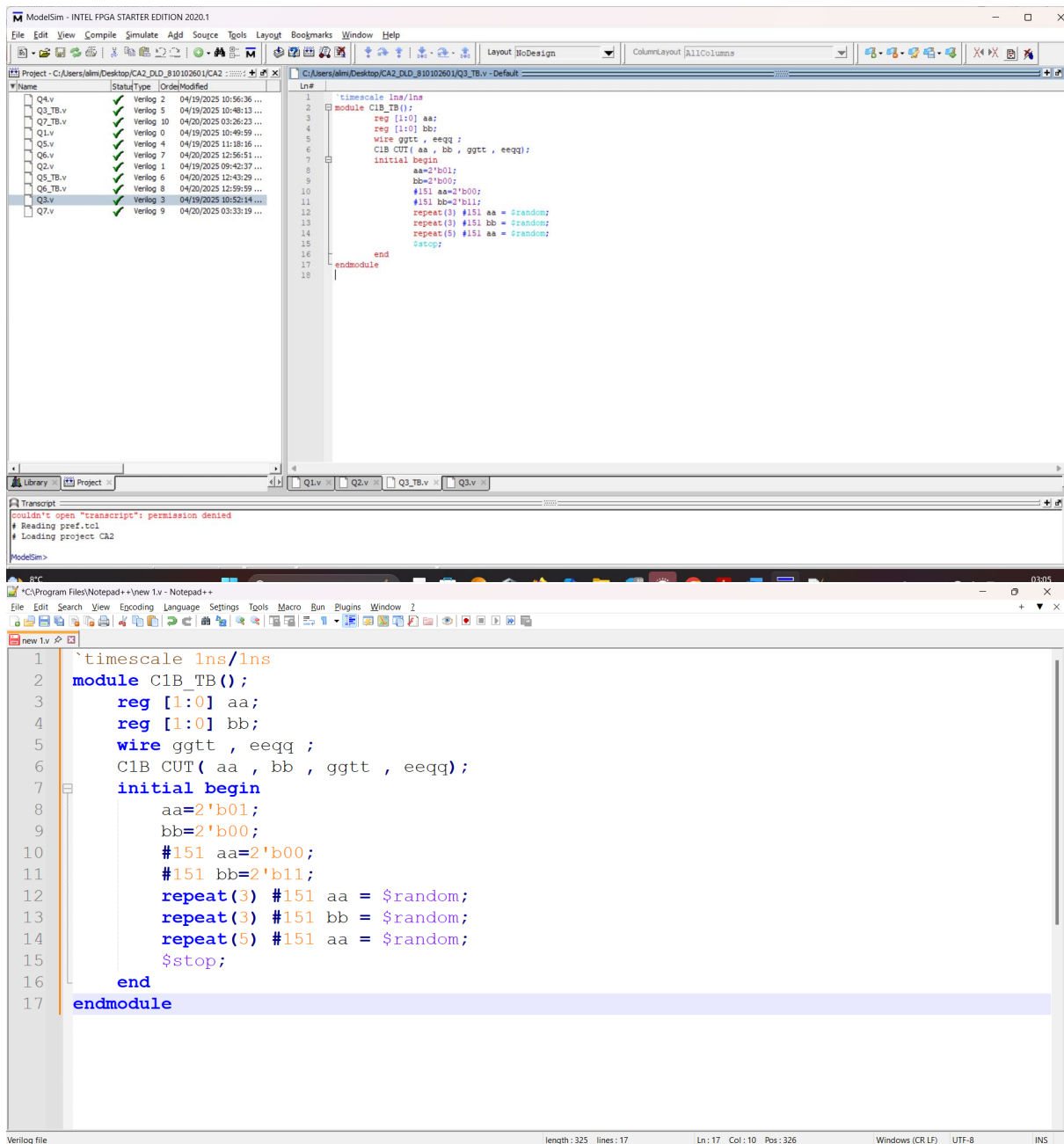
Q3)

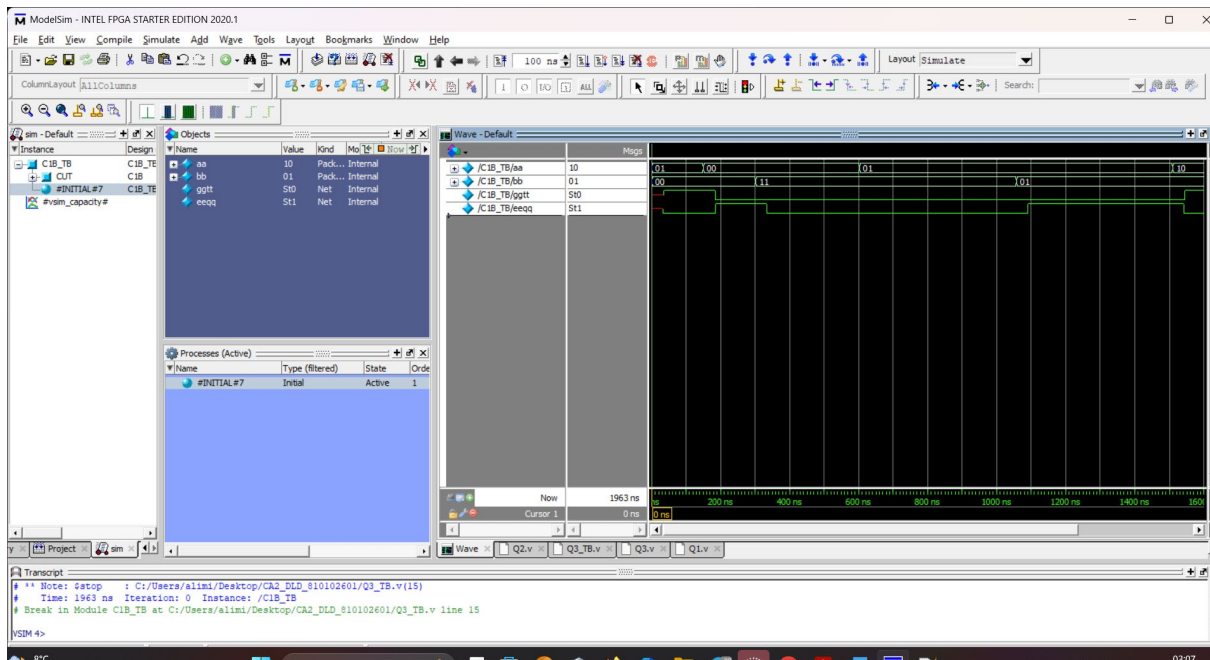


Verilog code:

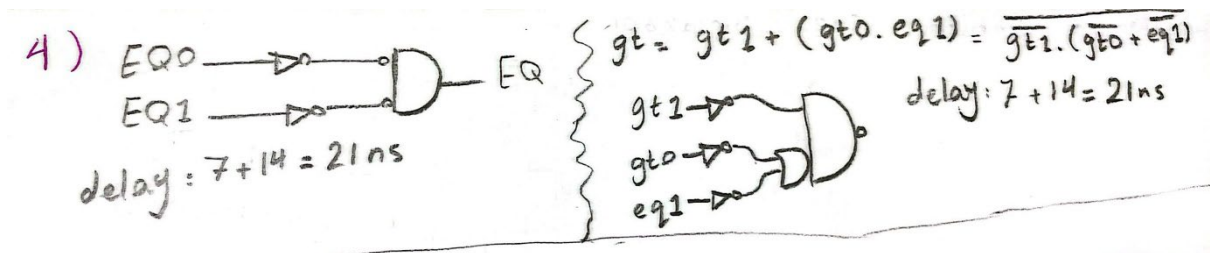


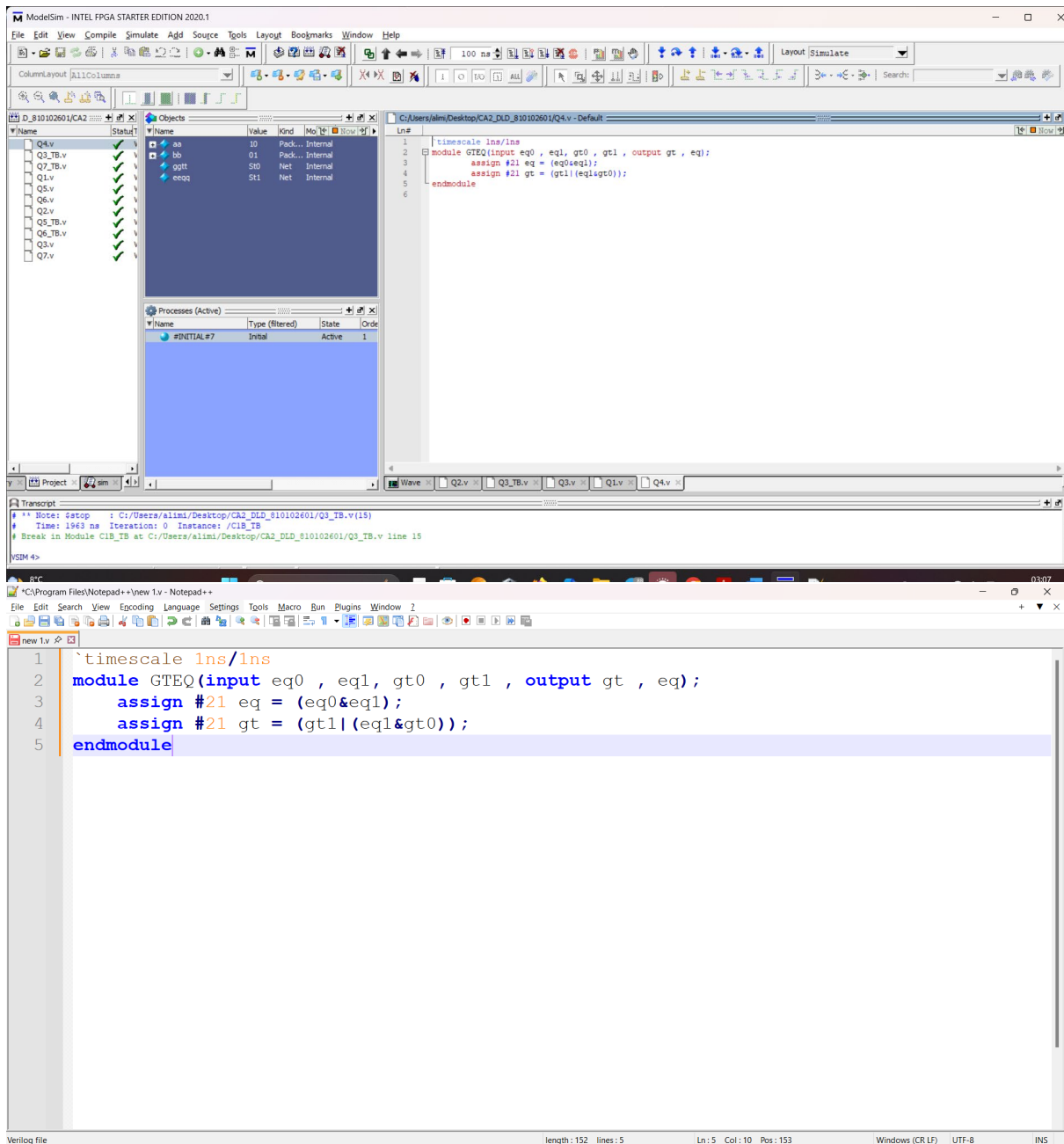
Testbench:



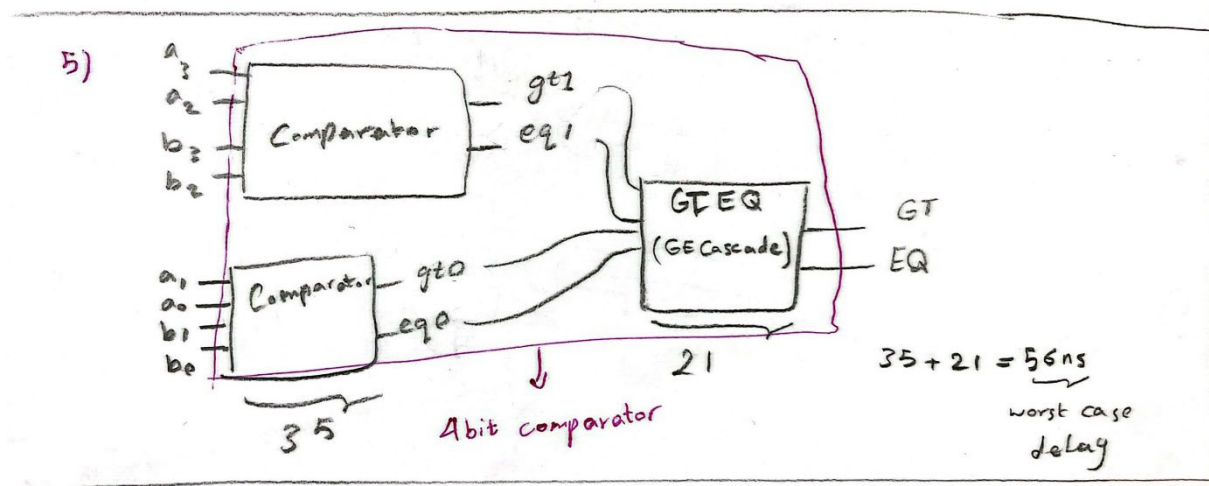


Q4)

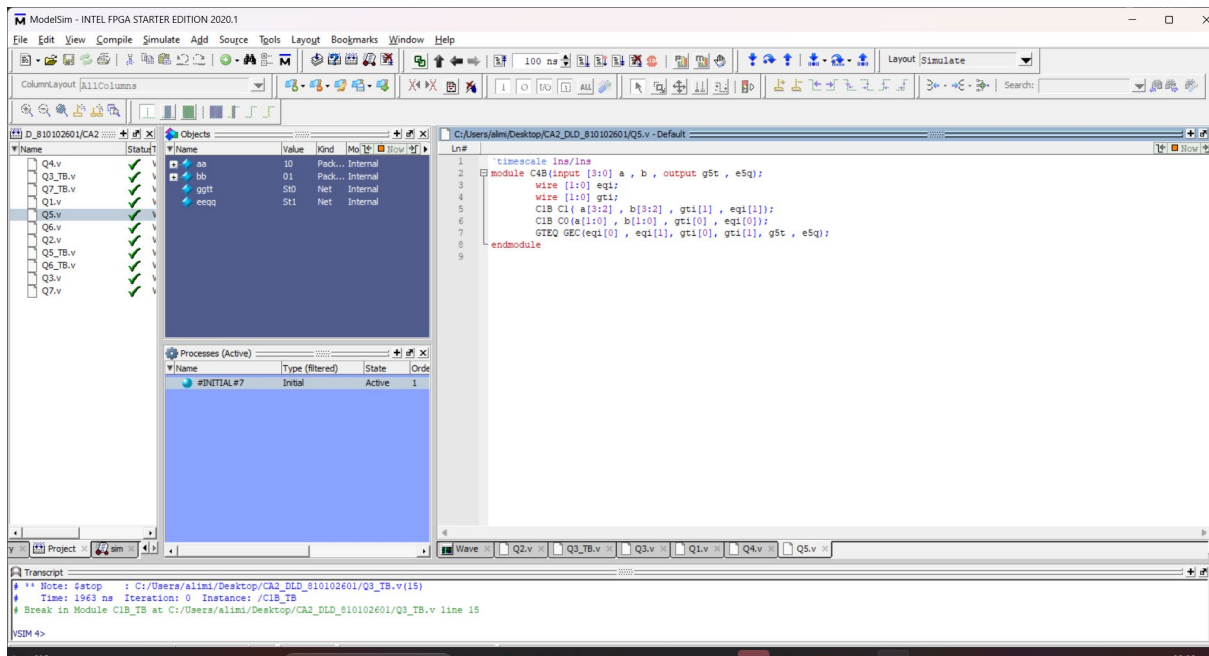




Q5)



VERILOG:



```
*C:\Program Files\Notepad++\new 1.v - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2
new 1.v
1 `timescale 1ns/1ns
2 module C4B(input [3:0] a , b , output g5t , e5q);
3     wire [1:0] eqi;
4     wire [1:0] gti;
5     C1B C1( a[3:2] , b[3:2] , gti[1] , eqi[1]);
6     C1B C0(a[1:0] , b[1:0] , gti[0] , eqi[0]);
7     GTEQ GEC(eqi[0] , eqi[1] , gti[0] , gti[1] , g5t , e5q);
8 endmodule

Verilog file length: 263 lines: 8 Ln: 8 Col: 10 Pos: 264 Windows (CR LF) UTF-8 INS
```

## TEST BENCH:

ModelSim - INTEL FPGA STARTER EDITION 2020.1

File Edit View Compile Simulate Add Source Tools Layout Bookmarks Window Help

ColumnLayout AllColumns

Objects

| Name    | Status | Name | Value | Kind    | Mo       | Row |
|---------|--------|------|-------|---------|----------|-----|
| Q4.v    | ✓      | aa   | 10    | Pack... | Internal |     |
| Q3_TB.v | ✓      | bb   | 01    | Pack... | Internal |     |
| Q7_TB.v | ✓      | S0   | S0    | Net     | Internal |     |
| Q1.v    | ✓      | ggt  | S1    | Net     | Internal |     |
| Q5.v    | ✓      | eeq  |       |         |          |     |
| Q6.v    | ✓      |      |       |         |          |     |
| Q5_TB.v | ✓      |      |       |         |          |     |
| Q6_TB.v | ✓      |      |       |         |          |     |
| Q3.v    | ✓      |      |       |         |          |     |
| Q7.v    | ✓      |      |       |         |          |     |

Processes (Active)

| Name       | Type (filtered) | State  | Order |
|------------|-----------------|--------|-------|
| #INITIAL#7 | Instal          | Active | 1     |

Ln#

```
1 `timescale 1ns/1ns
2 module C4B_TB();
3     reg [3:0] aa;
4     reg [3:0] bb;
5     wire g5t , ee5q;
6     C4B C1( aa , bb , g5t , ee5q);
7     initial begin
8         aa=4'b0001;
9         bb=4'b0000;
10        #151 aa=4'b0000;
11        #151 bb=4'b0101;
12        repeat(3) #151 aa = $random;
13        repeat(3) #151 bb = $random;
14        repeat(5) #151 aa = $random;
15        $stop;
16    end
17 endmodule
18
```

Transcript

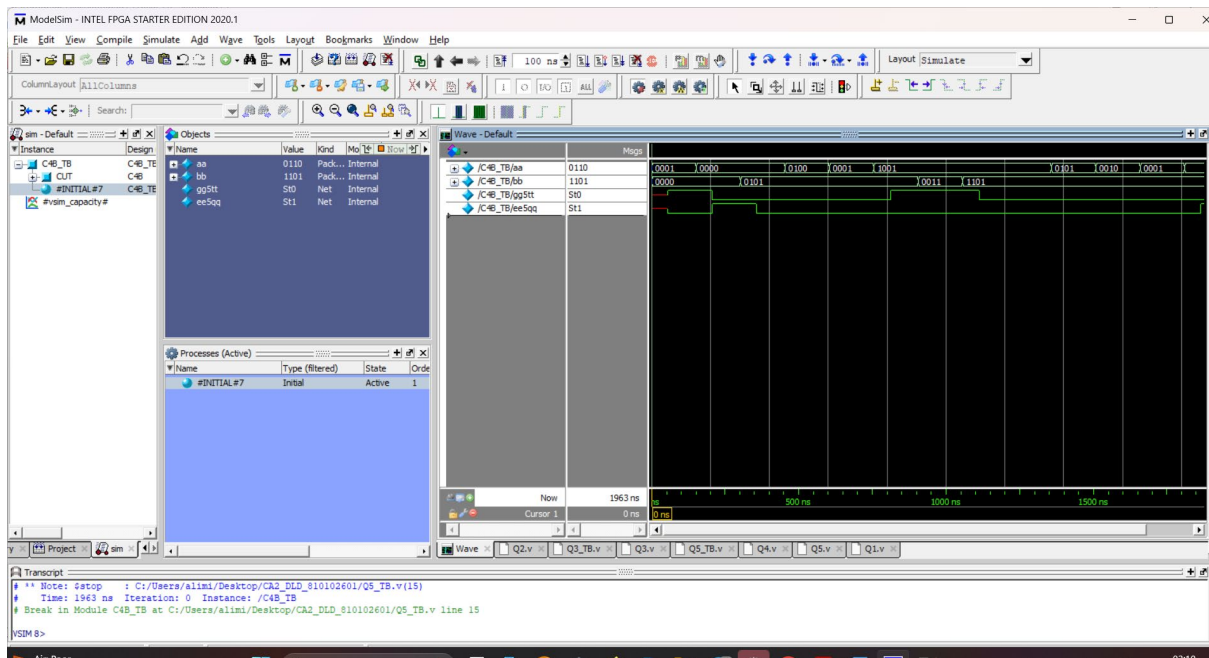
```
# ** Note: $stop : C:/Users/alimi/Desktop/CA2_DLD_810102601/Q3_TB.v(15)
# Time: 1963 ns Iteration: 0 Instance: /C1B_TB
# Break in Module C1B_TB at C:/Users/alimi/Desktop/CA2_DLD_810102601/Q3_TB.v line 15
VSIM 4>
```

```

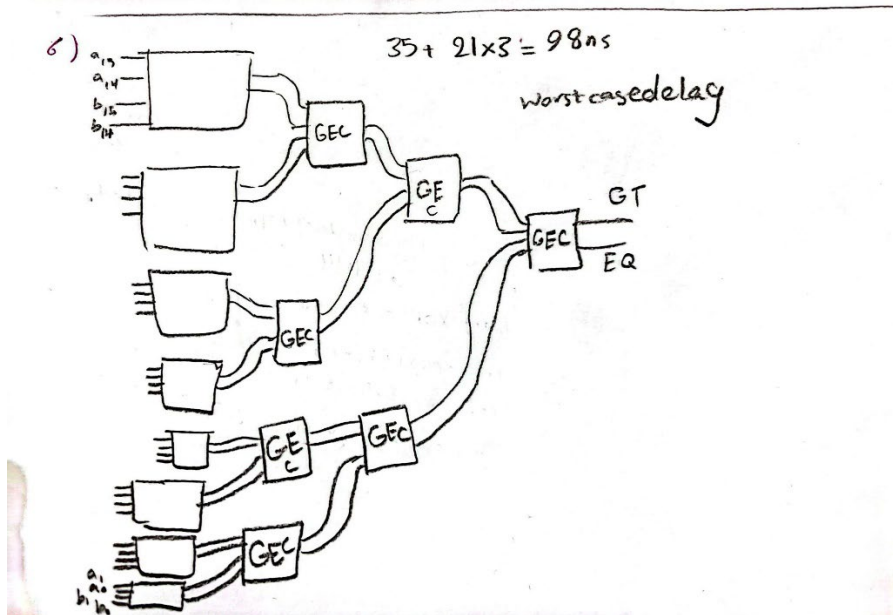
1  `timescale 1ns/1ns
2  module C4B_TB();
3      reg [3:0] aa;
4      reg [3:0] bb;
5      wire gg5tt , ee5qq ;
6      C4B CUT( aa , bb , gg5tt , ee5qq);
7      initial begin
8          aa=4'b0001;
9          bb=4'b0000;
10         #151 aa=4'b0000;
11         #151 bb=4'b0101;
12         repeat(3) #151 aa = $random;
13         repeat(3) #151 bb = $random;
14         repeat(5) #151 aa = $random;
15         $stop;
16     end
17 endmodule

```

Verilog file      length: 337   lines: 17      Ln: 17   Col: 10   Pos: 338      Windows (CR LF)   UTF-8      INS



Q6)



## VERILOG

```

module C4B_TB(input [15:0] a, b, output g6t, e6t);
    wire [7:0] eqj;
    wire [7:0] gtj;
    wire [2:0] eq_layer1;
    wire [2:0] gt_layer1;
    wire [2:0] eq_layer2;
    wire [2:0] gt_layer2;

    generate
    for (j = 0; j < 8; j = j + 1) begin
        CIB cmp2bit(a[2*j+1 : 2*j], b[2*j+1 : 2*j], gtj[j], eqj[j]);
    end
    endgenerate

    genvar k;
    generate
    for (k = 0; k < 4; k = k + 1) begin
        GTEQ ge_layer1(eqj[2*k], eqj[2*k+1], gtj[2*k], gtj[2*k+1], gt_layer1[k], eq_layer1[k]);
    end
    endgenerate

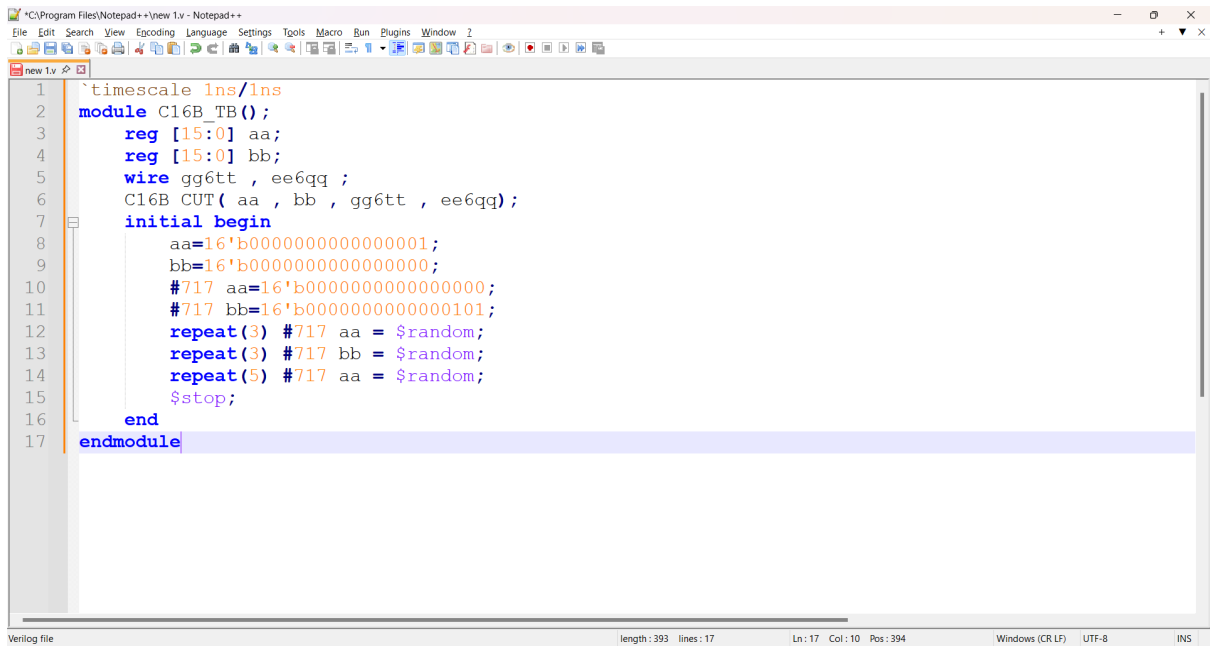
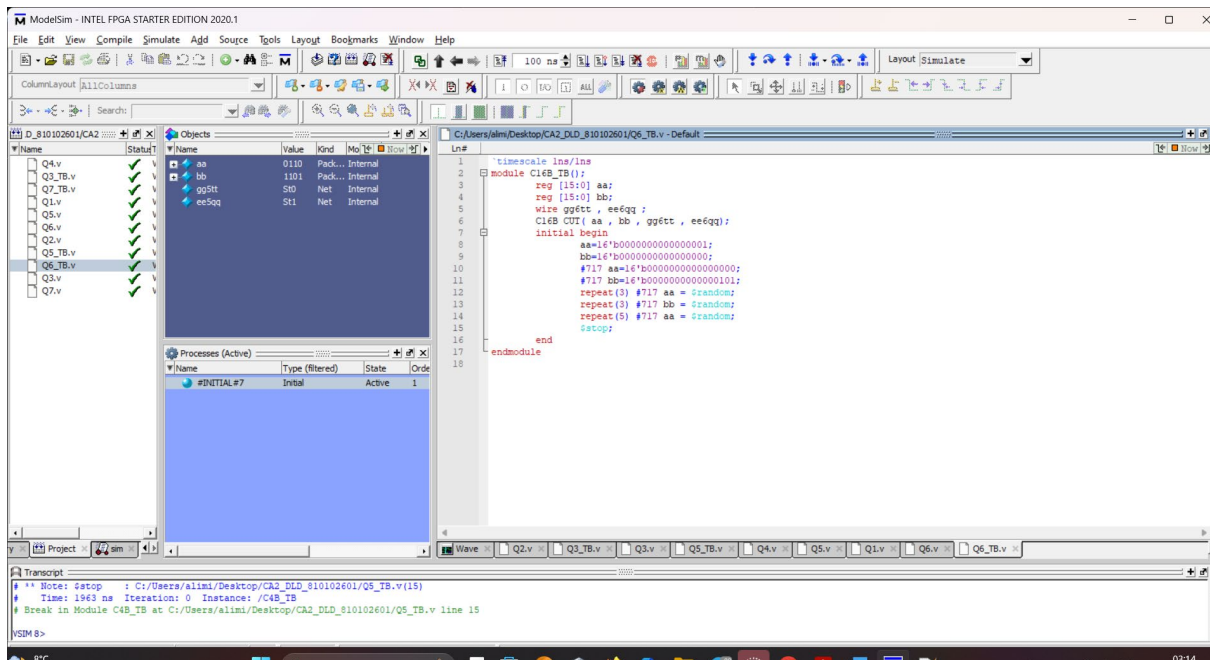
    generate
    for (k = 0; k < 2; k = k + 1) begin
        GTEQ ge_layer2(eq_layer1[2*k], eq_layer1[2*k+1], gt_layer1[2*k], gt_layer1[2*k+1], gt_layer2[k], eq_layer2[k]);
    end
    endgenerate

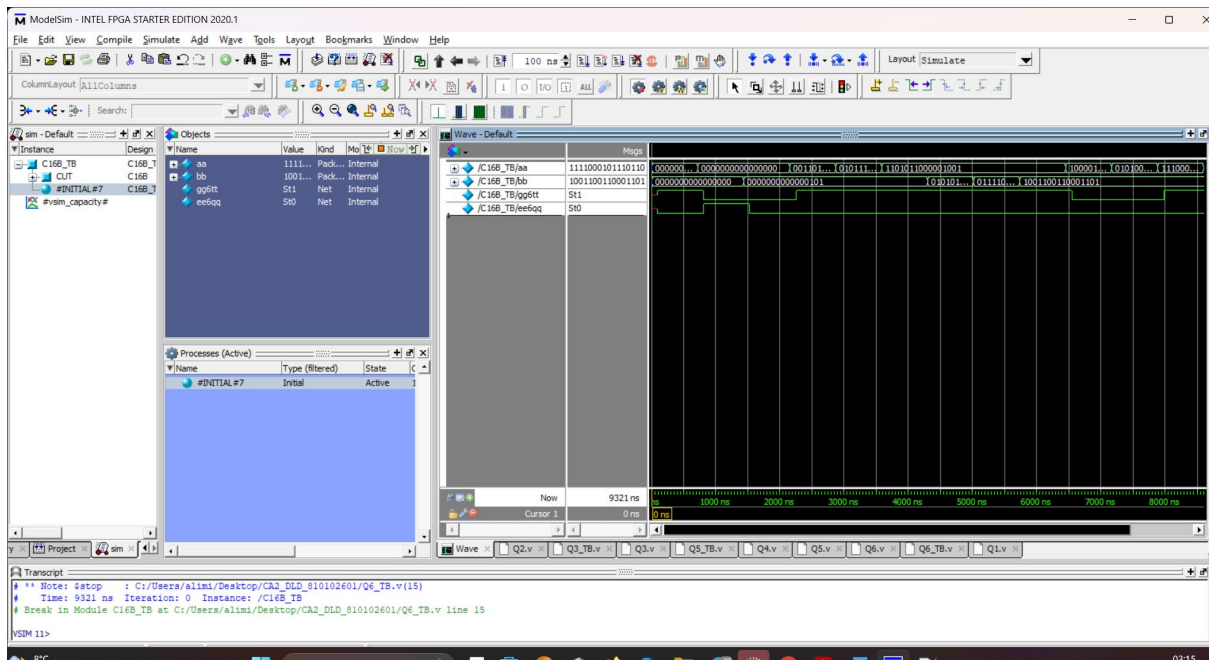
    GTEQ final_compare(eq_layer2[0], eq_layer2[1], gt_layer2[0], gt_layer2[1], g6t, e6t);
endmodule

```

Verilog file: length: 981 lines: 32 Ln: 32 Col: 1 Pos: 982 Windows (CR LF) UTF-8 INS

TEST BENCH:





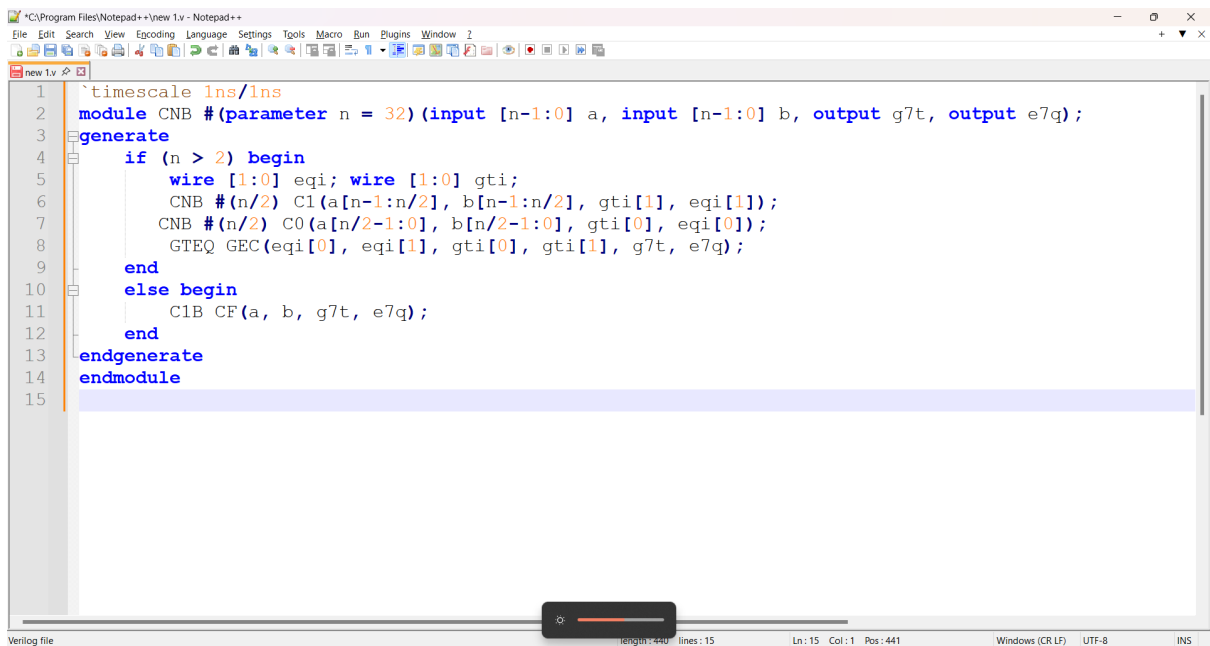
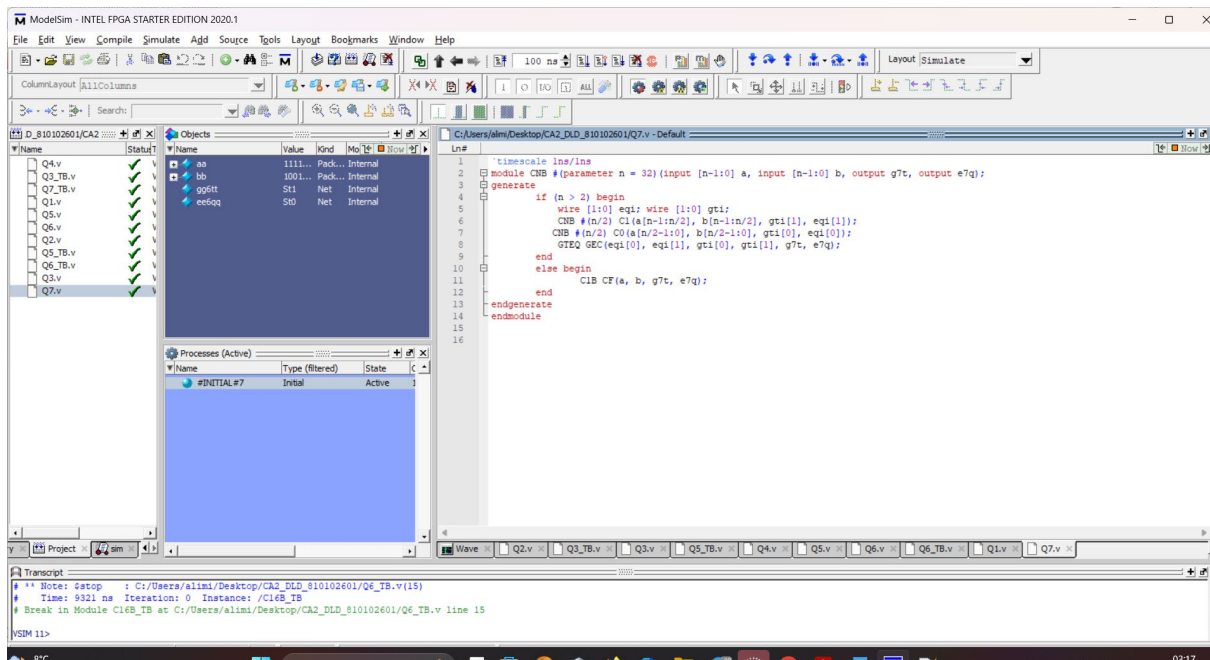
Q7)

IN THIS QUESTION, we used recursive function algorithm. (because we didn't know I or N) and we at end of the recursive function we used 1 bit comparator (that we built in Q3).

To calculate worst case delay: we must consider one 2 bit comparator and (i-1) GE cascades.

$$7) \quad \underbrace{35}_{\text{Comp}} + (i-1) \times \underbrace{21}_{\text{GE cascade}} = \text{worst case delay (ns)}$$

VERILOG:



Test bench:

ModelSim - INTEL FPGA STARTER EDITION 2020.1

File Edit View Compile Simulate Add Source Tools Layout Bookmarks Window Help

ColumnLayout AllColumns

Search:

Objects

| Name    | Status | Name  | Value   | Kind    | Mo       | Te | How | FF |
|---------|--------|-------|---------|---------|----------|----|-----|----|
| Q4.v    | ✓      | aa    | 1111... | Pack... | Internal |    |     |    |
| Q3_TB.v | ✓      | bb    | 1001... | Pack... | Internal |    |     |    |
| Q7_TB.v | ✓      | gg7tt | S11     | Net     | Internal |    |     |    |
| Q1.v    | ✓      | ee7qq | S10     | Net     | Internal |    |     |    |
| Q5.v    | ✓      |       |         |         |          |    |     |    |
| Q6.v    | ✓      |       |         |         |          |    |     |    |
| Q2.v    | ✓      |       |         |         |          |    |     |    |
| Q5_TB.v | ✓      |       |         |         |          |    |     |    |
| Q6_TB.v | ✓      |       |         |         |          |    |     |    |
| Q3.v    | ✓      |       |         |         |          |    |     |    |
| Q7.v    | ✓      |       |         |         |          |    |     |    |

Processes (Active)

| Name       | Type (Filtered) | State  | C |
|------------|-----------------|--------|---|
| #INITIAL#7 | Initial         | Active |   |

Ln#

```

1 | timescale 1ns/1ns
2 | module CNB_TB();
3 |     parameter nn = 32;
4 |     reg [nn-1:0] aa;
5 |     reg [nn-1:0] bb;
6 |     wire gg7tt, ee7qq;
7 |     CNB CUT aa, bb, gg7tt, ee7qq;
8 |     initial begin
9 |         aa=1;
10 |        bb=0;
11 |        #1103 bb=1;
12 |        repeat(3) #1103 aa = $random;
13 |        repeat(3) #1103 bb = $random;
14 |        repeat(5) #1103 aa = $random;
15 |        $stop;
16 |    end
17 | endmodule
18 |

```

Transcript

```

** Note: $stop : C:/Users/alimi/Desktop/CA2_DLD_810102601/Q6_TB.v(15)
* Time: 8321 ns Iteration: 0 Instance: /C168_TB
* Break in Module C168_TB at C:/Users/alimi/Desktop/CA2_DLD_810102601/Q6_TB.v line 15
VCSM 11>

```

\*CAProgram Files\notepad++\new 1.v - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window 2

new 1.v

```

4 | if (n > 2) begin
5 |     wire [1:0] eqi; wire [1:0] gti;
6 |     CNB #(n/2) C1(a[n-1:n/2], b[n-1:n/2], gti[1], eqi[1]);
7 |     CNB #(n/2) C0(a[n/2-1:0], b[n/2-1:0], gti[0], eqi[0]);
8 |     GTEQ GEC(eqi[0], eqi[1], gti[0], gti[1], g7t, e7q);
9 | end
10 | else begin
11 |     C1B CF(a, b, g7t, e7q);
12 | end
13 | endgenerate
14 | endmodule
15 | timescale 1ns/1ns
16 | module CNB_TB();
17 |     parameter nn = 32;
18 |     reg [nn-1:0] aa;
19 |     reg [nn-1:0] bb;
20 |     wire gg7tt, ee7qq;
21 |     CNB CUT aa, bb, gg7tt, ee7qq;
22 |     initial begin
23 |         aa=1;
24 |         bb=0;
25 |         #1103 bb=1;
26 |         repeat(3) #1103 aa = $random;
27 |         repeat(3) #1103 bb = $random;
28 |         repeat(5) #1103 aa = $random;
29 |         $stop;
30 |     end
31 | endmodule

```

Verilog file

length: 770 lines: 31 Ln: 31 Col: 10 Pos: 771 Windows (CR LF) UTF-8 INS

